

Embodied AI in Autonomous Driving Final Project

Retrieve-then-Rerank Inference for Vocabulary-Based End-to-End Driving

Hyunjun Kim
Donggue Kim
Jeongwoo Park



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Introduction

1. E2E AD path panning
2. Problems of Vocabulary-based planning

End-to-End Autonomous Driving Path Planning

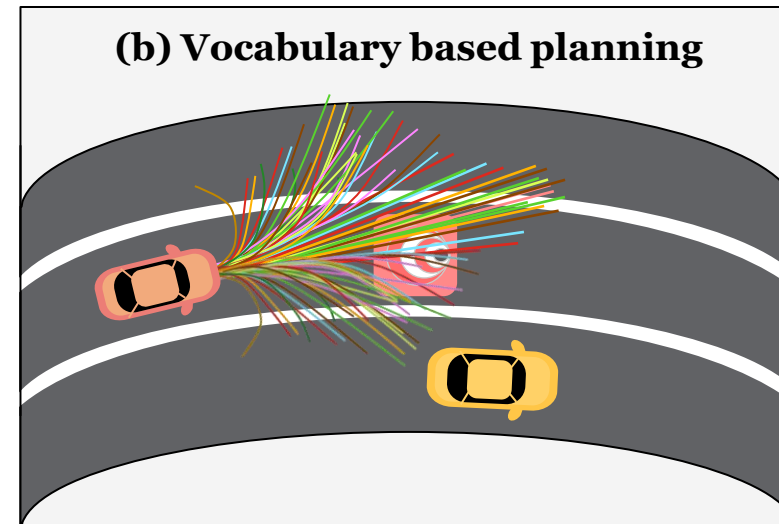
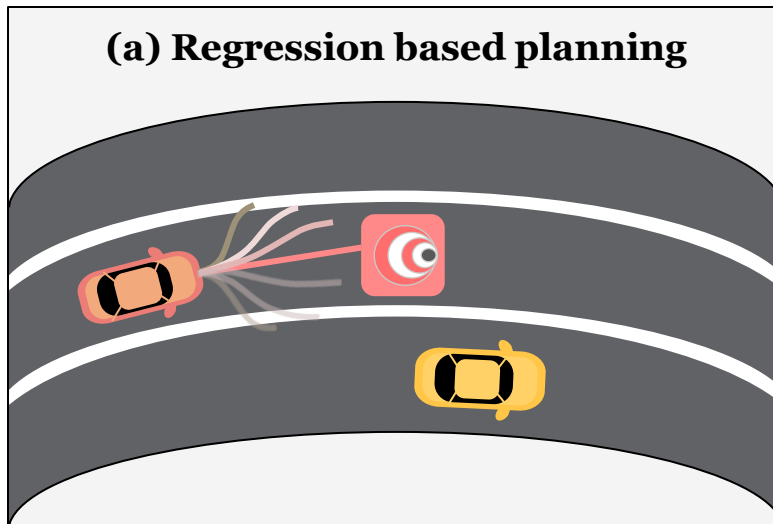
Approaches

► 1. Regression based planning

- Directly regress a single future trajectory from the current scene
 - ex. VAD, GenAD

► 2. Vocabulary based planning

- Select and score candidates from a fixed trajectory vocabulary
 - Hydra-MDP, VADv2



Path planning approaches in E2E AD

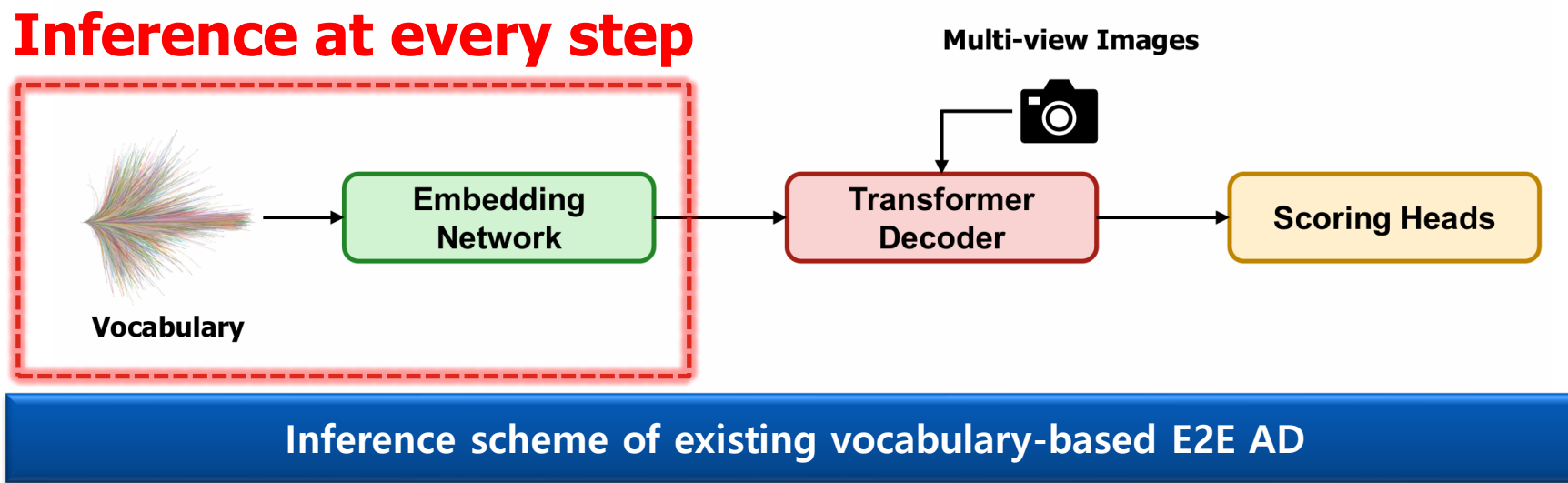
Vocabulary based Planning Overview

■ Methods

- ▶ Forward the fixed vocabulary embeddings at every step

■ Problem

- ▶ As the reusable fixed candidates, the vocabulary embeddings, are inferred at every step, latency surges as size and complexity increase



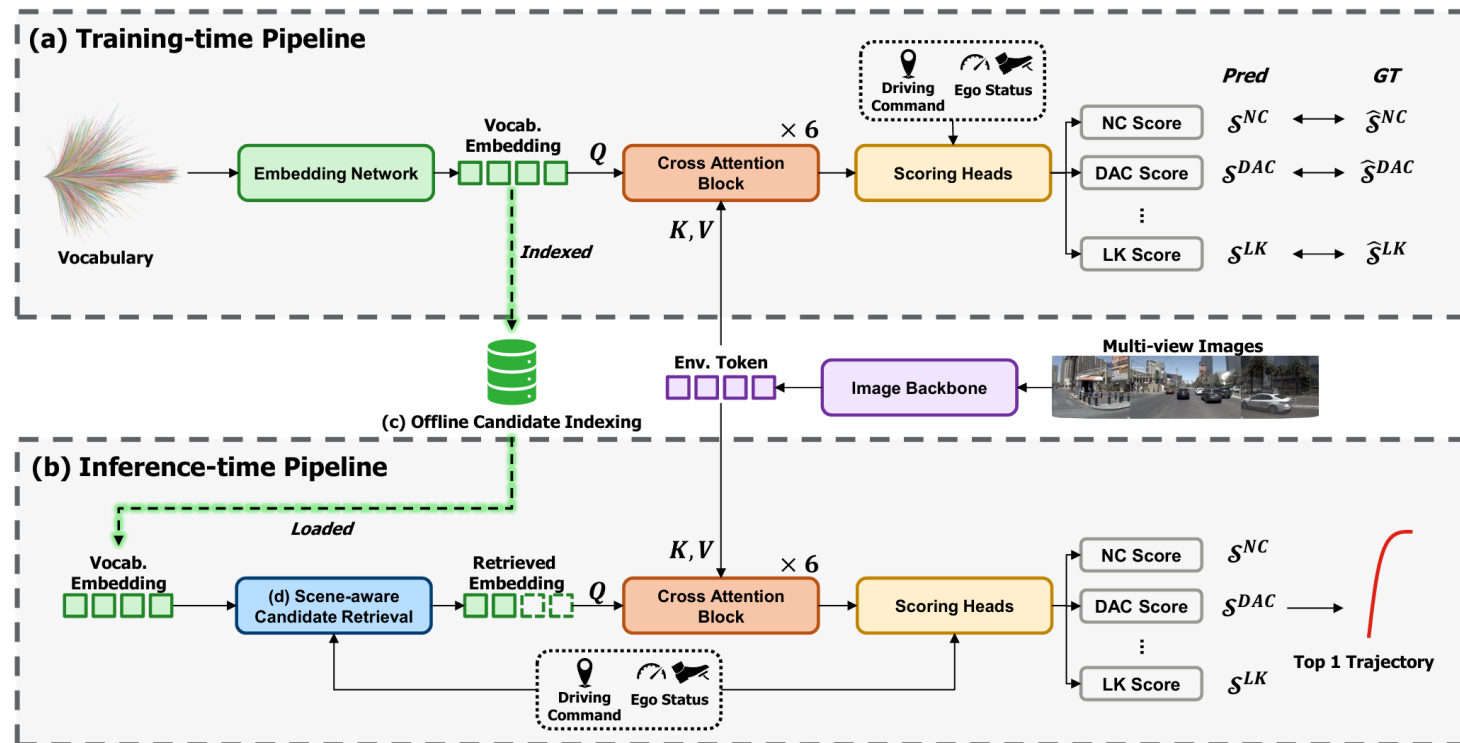
Proposed Solution: Retrieve-then-Rerank

Phase1. Training

- ▶ Input: (Vocabulary), (Multi-view Images), (ego status), (driving command)
- ▶ Output: (Cached vocabulary embedding), (Score)

Phase2. Inference

- ▶ Input : (Multi-view Images), (ego status), (driving command)
- ▶ Output : (Score), (Top-1 Trajectory)



Methods

1. OCI (Offline Candidate Indexing)
2. SCR (Scene-aware Candidate Retrieval)

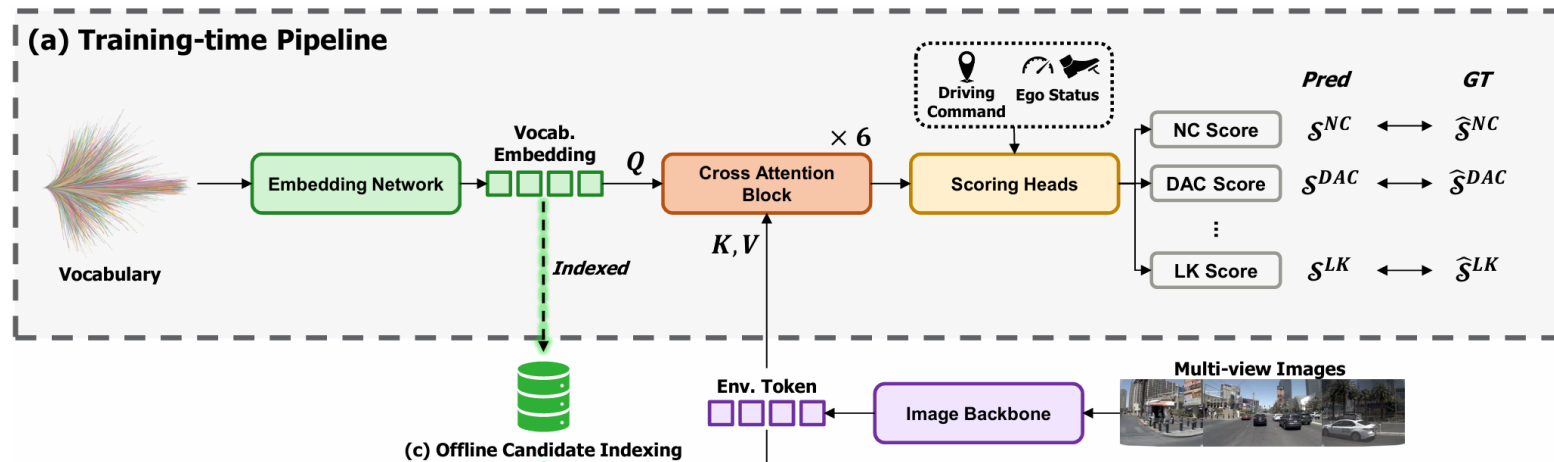
[Phase 1] Vocabulary based Planning Forward

■ 방법

- ▶ No retrieval is used during training → forward the entire Vocabulary as-is to learn the Score

■ Score 정리

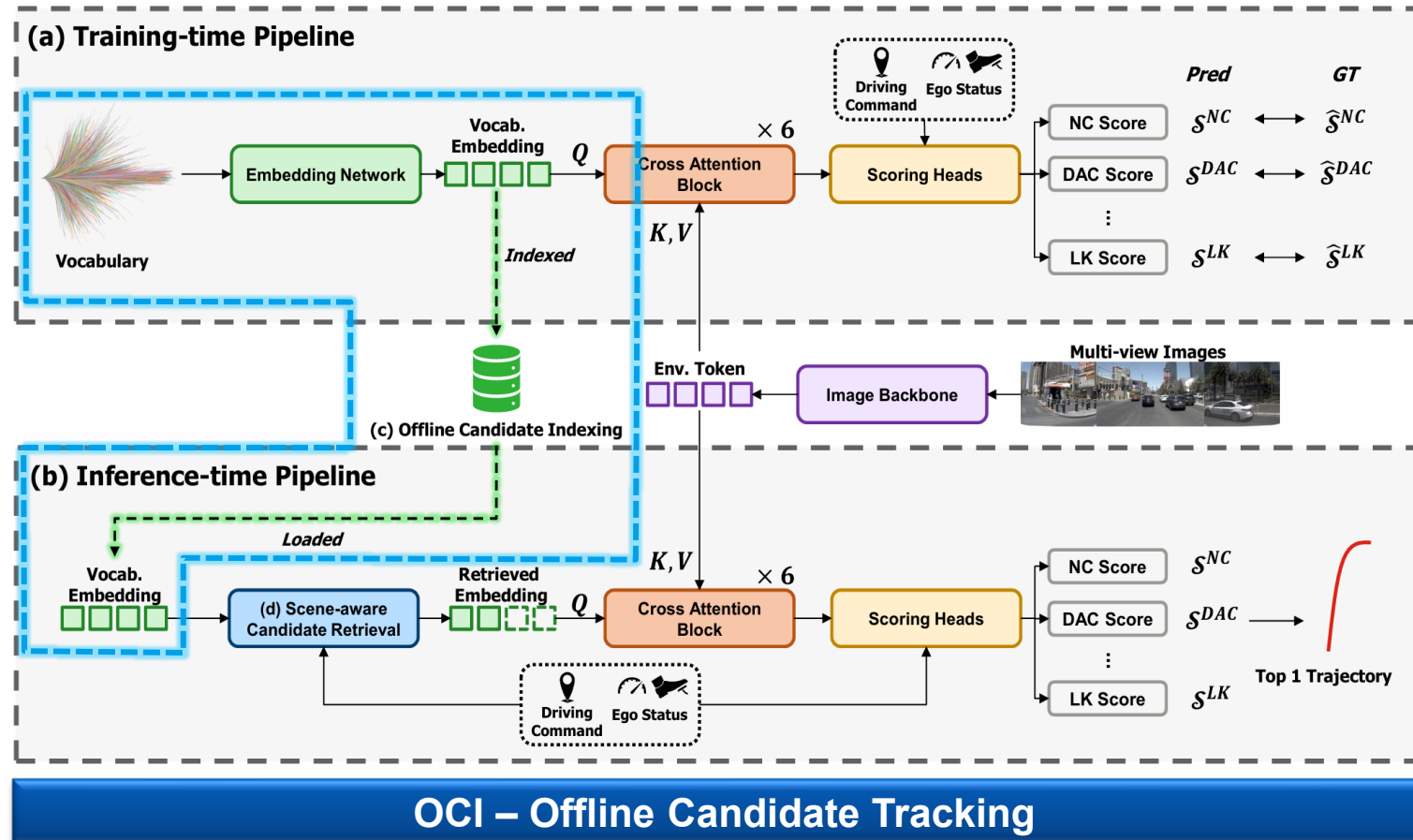
Score	설명	Score	설명
NC	No at fault collision	TTC	Time to collision within bound
DAC	Driving area compliance	LK	Lane keeping
TL	Traffic light compliance	HC	History comfort
EP	Ego progress	EC	Two frame and extended comfort



[Phase 2] OCI - Offline Candidate Indexing

■ 방법

- ▶ **Index** the scene-invariant candidate embeddings **only once offline** and load them at inference **without recomputation**



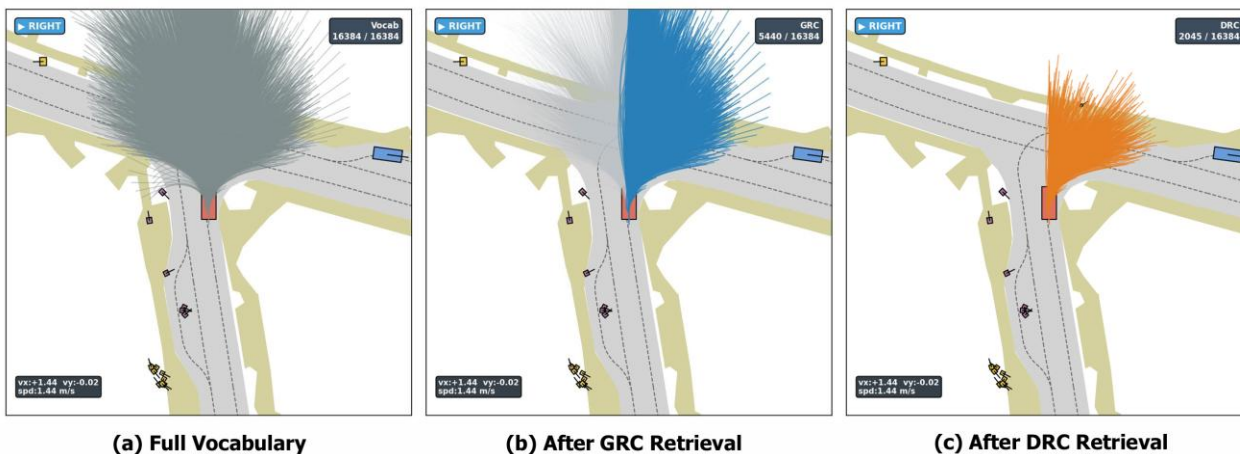
[Phase 2] SCR – Scene-aware Candidate Retrieval

Method

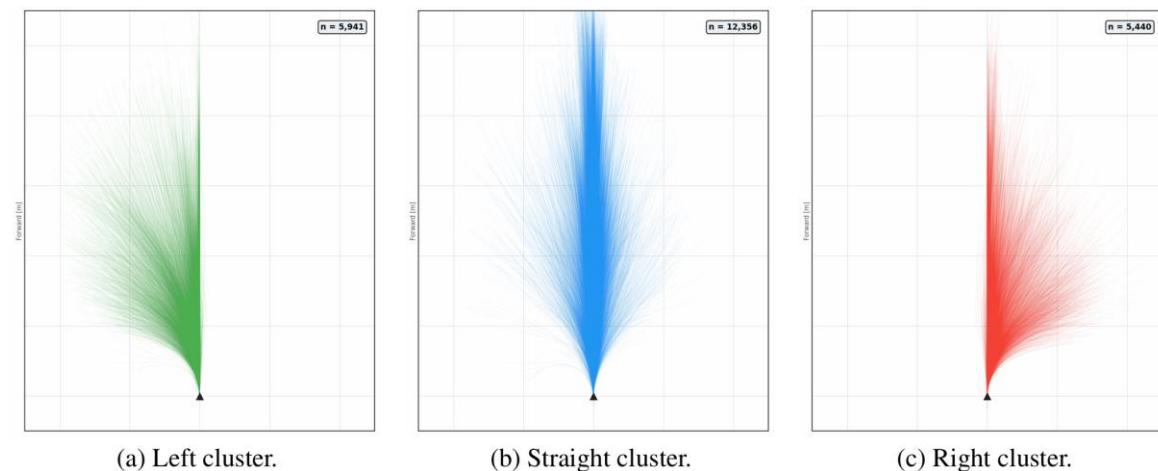
- ▶ A method to narrow the indexed Vocabulary in two stages before evaluation
- ▶ (Vocabulary) → [GRC] → [DRC] → [Candidates]

Score

- ▶ GRC – Global Route Compliance
 - Driving command (left turn / straight / right turn) → precomputed cluster lookup (cost ≈ 0)
- ▶ DRC – Dynamic Reachability Compliance
 - Dynamic reachability judgment based on ego velocity



SCR — Total 16,384 → GRC (right 5,440) → DRC 2,045



clusters by left/straight/right command — GRC

Experiments

Experiments (1/4)

■ Training and inference environment

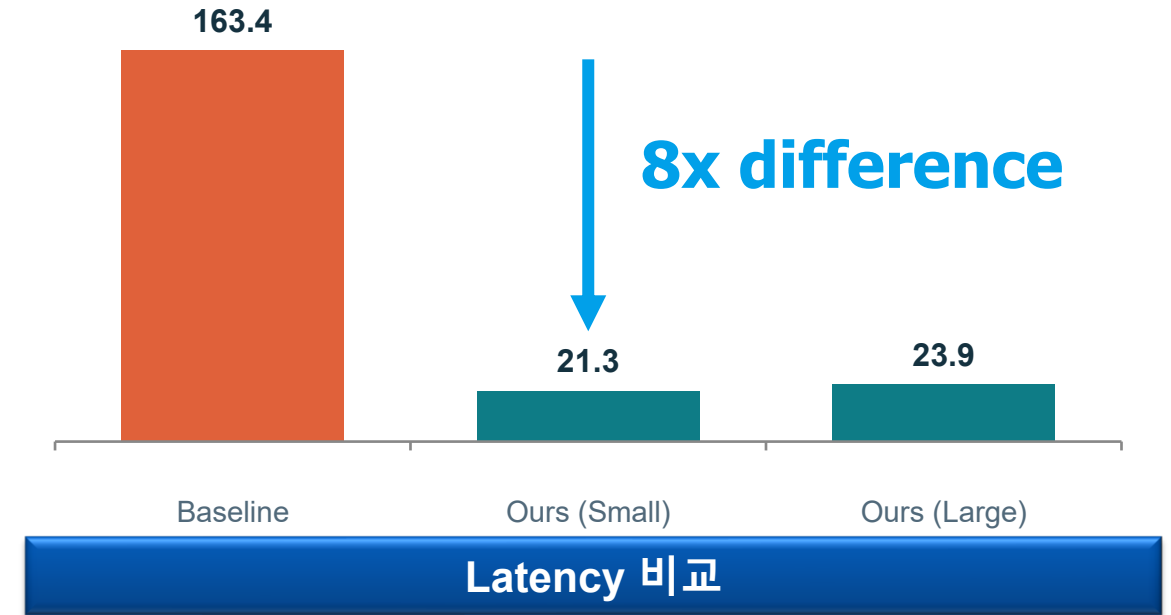
- ▶ 4x RTX 4090
- ▶ batch size: 1

■ Dataset

- ▶ NAVSIM(navtrain/ navtest) + navhard

■ Metric

- ▶ EPDMS
 - NAVSIM Benchmark comprehensive evaluation metric



Method	Vocab.	Backbone	Latency (ms)↓	NAVSIM v2 navhard			NAVSIM v2 navtest
	Size			EPDMS-1↑	EPDMS-2↑	EPDMS↑	EPDMS↑
Baseline	16,384	V2-99	163.4	74.1	48.9	37.6	86.9
Ours (Small)	8,192	V2-99	21.3	74.3	49.0	38.0	85.3
Ours (Large)	16,384	V2-99	<u>23.9</u>	75.2	50.1	40.2	86.1

Quantitative evaluation results – NAVSIM v2 navhard / navtest

Experiments (2/4)

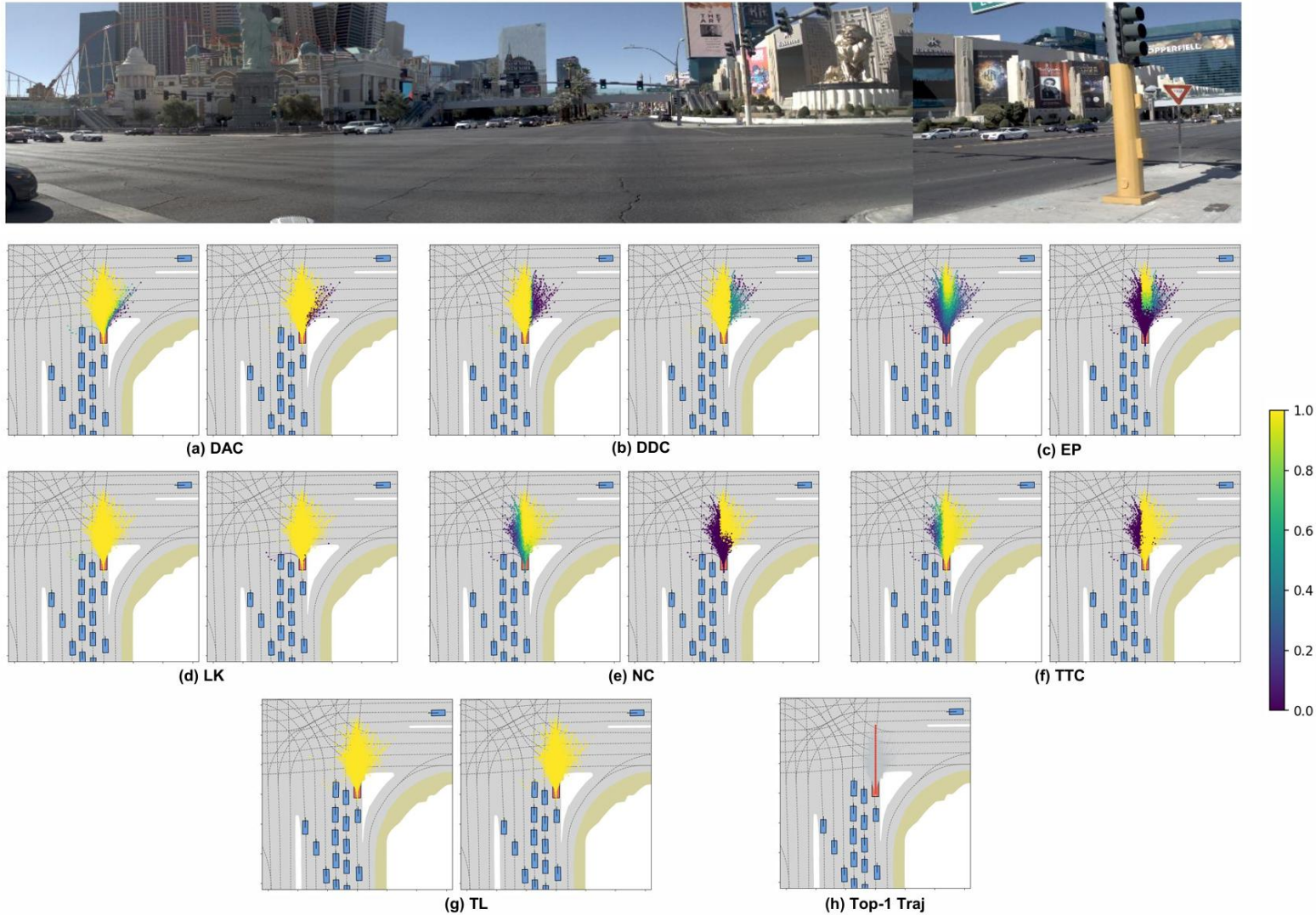
■ NAVSIM v2 – navtest evaluation results

- Slight performance drop versus baseline, **but 8× faster in terms of latency**

Method	Backbone	NC↑	DAC↑	DDC↑	TL↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑	EPDMS _{HF} ↑
Ego Status MLP	ResNet-34	93.1	77.9	92.7	99.6	86.0	91.5	89.4	98.3	85.4	64.0	–
TransFuser [4]	ResNet-34	96.9	89.9	97.8	99.7	87.1	95.4	92.7	98.3	87.2	76.7	–
DriveSuprim [27]	ResNet-34	97.5	96.5	99.4	99.6	88.4	96.6	95.5	98.3	77.0	83.1	–
DiffusionDriveV2 [29]	ResNet-34	97.7	96.6	99.2	99.8	88.3	97.2	96.0	97.8	91.0	85.5	–
ARTEMIS [7]	ResNet-34	98.3	95.1	98.6	99.8	81.5	97.4	96.5	98.3	98.3	83.1	–
Hydra-MDP++ [14]	V2-99	98.4	98.0	99.4	99.8	87.5	97.7	95.3	98.3	77.4	85.1	–
Baseline	V2-99	98.5	98.7	99.6	99.9	88.4	98.2	97.0	98.3	79.4	86.9	90.5
Ours (Small)	V2-99	98.1	98.0	99.4	99.9	87.8	97.5	96.3	98.2	77.9	85.3	88.7
Ours (Large)	V2-99	98.5	98.0	99.6	99.8	89.1	98.1	96.4	98.3	77.5	86.1	89.6

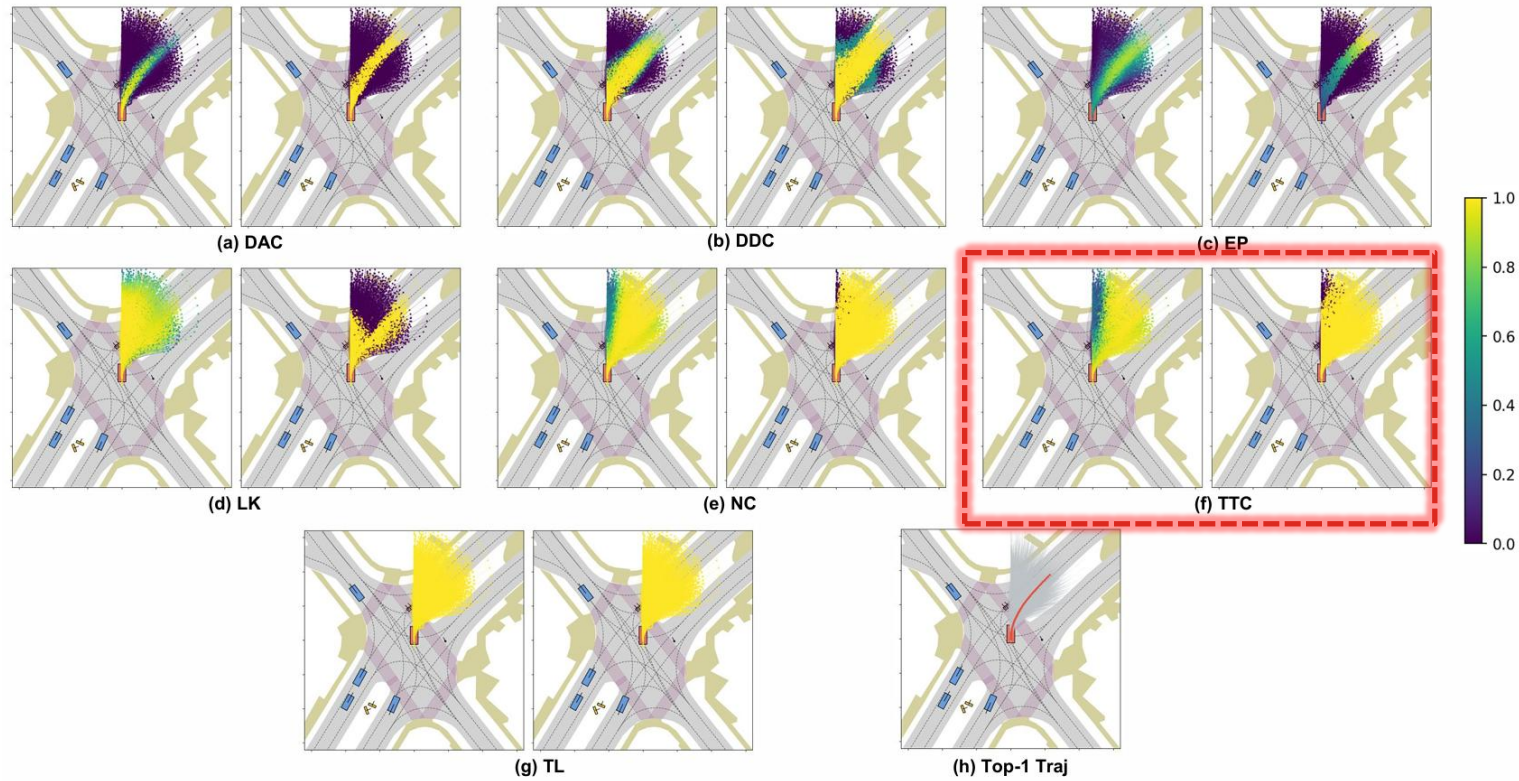
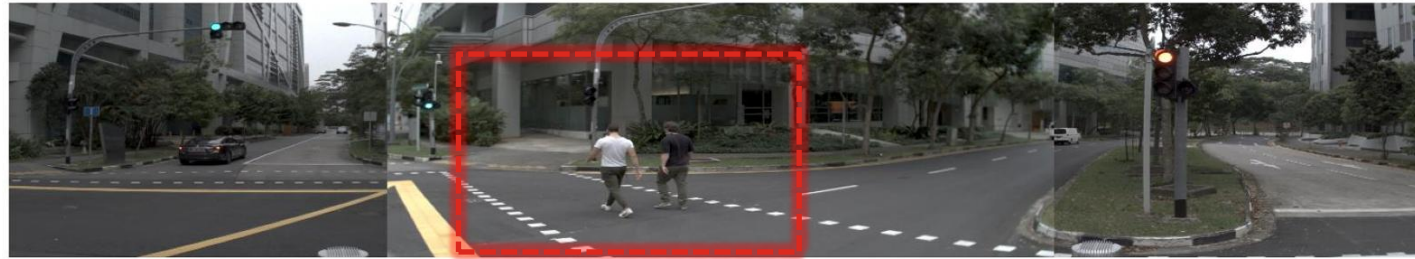
Quantitative evaluation results– NAVSIMv2 navtest

Experiments (3/4) – Straight Case



Qualitative Result – Straight Case

Experiments (4/4) – Turning Case



Qualitative Result – Turning Case

Conclusion



Conclusion

■ Retrieve-then-rerank framework

- ▶ Propose a method to quickly retrieve and rerank the existing fixed Vocabulary using OCI+SCR

■ Efficiency–quality trade-off improvement

- ▶ 163.4ms → 23.9ms ($\approx 6.8x$), navtest EPDMS 86.1 maintained, navhard 37.6 → 40.2 improved)



**THANK YOU
FOR YOUR ATTENTION**